

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

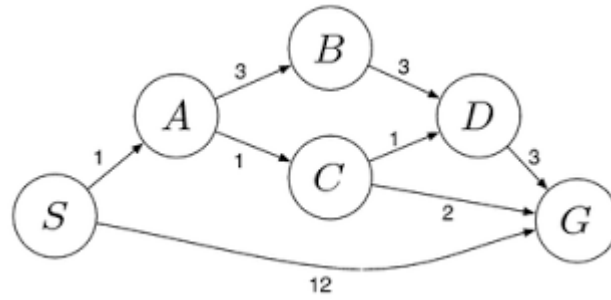
Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Q1. Water Jug Problem (4-Gallon and 3-Gallon Jug)

Problem Understanding

Given:

- Jug A = 4 gallons
- Jug B = 3 gallons
-

Goal:

Obtain exactly **2 gallons in the 4-gallon jug**.

Method Used

Use **State Space Search**.

Represent each state as:

(4-gallon jug 3-gallon jug)

Initial state:

(0, 0)

Goal state:

(2, 0)

Solution Process

Step	Action	State (4G,3G)
1	Fill 3-gallon jug	(0,3)
2	Pour into 4-gallon jug	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4-gallon jug	(2,0)

Final Answer

The required 2 gallons are obtained in the 4-gallon jug.

Interpretation

The Water Jug problem is a classical AI State Space Search problem, where each state represents the amount of water in both jugs and operators generate successor states until the goal is reached.

Q2. Mars Rover Intelligent Agent

(i) Percepts

The Mars Rover receives information from:

- Cameras
- Rock and soil sensors

- Temperature sensors
- Spectrometer
- GPS/Navigation sensors
- Obstacle sensors
- Battery level
- Communication signals

(ii) Operating Environment

PEAS Analysis

Feature	Description
Observable	Partially Observable
Agents	Single Agent
Deterministic	Stochastic
Episodes	Sequential
Environment	Dynamic
States	Continuous
Knowledge	Unknown

(iii) Actions

The rover can:

- Move forward
- Move backward
- Turn left
- Turn right
- Capture images
- Drill rocks
- Collect samples
- Analyze soil
- Send data to Earth
- Avoid obstacles
- Recharge (solar)

(iv) Performance Measure

The rover is evaluated using:

- Number of samples collected
- Accuracy of scientific analysis
- Distance covered
- Energy efficiency
- Obstacle avoidance
- Mission completion rate
- Communication success

(v) Suitable Agent Architecture

Learning Agent

Reason

A Learning Agent is suitable because it:

- Learns from previous exploration
- Improves navigation

- Avoids repeated mistakes
- Adapts to unknown terrain
- Optimizes battery usage

Interpretation

Mars Rover operates in a **dynamic and partially observable environment**, therefore a **Learning Agent** provides the best performance.

Q3. Formulation of 8-Queens Problem

Problem Understanding

Place **8 queens** on an **8×8 chessboard** such that:

- No two queens share the same row
- No two queens share the same column
- No two queens share the same diagonal

Problem Formulation

Initial State

Empty chessboard

State Space

Every possible arrangement of queens.

Maximum possible states:

$$8^8 = 16,777,216$$

Actions

Place one queen in a safe position.

Transition Model

Move from one valid configuration to another after placing a queen.

Goal Test

Exactly **8 queens** placed without attacking one another.

Path Cost

Each queen placement = 1

Minimum cost = 8 placements

Search Strategy

Use:

- Backtracking Search
- Depth First Search (DFS)

because they eliminate invalid branches early.

Interpretation

The 8-Queens problem is a **Constraint Satisfaction Problem (CSP)** where constraints eliminate invalid queen placements.

Q4. OLA Cab Booking Agent

Problem Understanding

The customer chooses:

- Pickup
- Destination
- Cab Type
- Booking

The system automatically selects the best cab.

Suitable Agent

Goal-Based Problem-Solving Agent

Reason

The goal is:

Reach the destination safely with the selected cab.

The agent evaluates available options before selecting the best action.

Pseudocode

START

Input Pickup Location

Input Destination

Display Available Cab Types

Mini

Micro

Sedan

Prime

Shared

User selects Cab

Search nearby drivers

IF Driver Available THEN

Assign Driver

Calculate Fare

Display Driver Details

Track Cab

Reach Destination

Generate Bill

ELSE

Display "No Cab Available"

ENDIF

STOP

Interpretation

Goal-Based Agents evaluate available actions before choosing the one that satisfies the user's travel goal.

Q5. Uniform Cost Search (UCS)

Problem Understanding

Objective:

Find the **minimum-cost path** from

$$S \rightarrow G$$

using **Uniform Cost Search (UCS)**.

Method

Uniform Cost Search always expands the node with the **lowest cumulative path cost**.

Algorithm:

1. Start from source.
2. Insert into Priority Queue.
3. Remove lowest-cost node.
4. Expand successors.
5. Repeat until Goal.

UCS Algorithm

UCS(Start, Goal)

Priority Queue $\leftarrow$ Start

while Queue not empty

Remove Lowest Cost Node

If Goal Found

Return Path

Else

Expand Children

Insert with Total Cost

End While

Interpretation

Uniform Cost Search guarantees the **optimal (least-cost) path** because it always expands the node with the smallest cumulative cost.